MARS ROUTE PLANNER

Data files, structure and AI use

A practical guide to what is in the dataset folder, what each product contains, and how it becomes route-planning inputs.

Current snapshot
The source folder contains global terrain, crater, geology, dune, thermal-inertia and AI4Mars products, plus rover observation samples. The full MCD 6.2 archive is now present as a 4.93 GB tar.gz but has not yet been extracted or validated.

What the system will do
Use map layers and engineering rules to calculate cell risks; use A* search to find candidate routes; optionally train a vision model on AI4Mars images to classify terrain. It will not learn all routing physics by feeding every archive directly into one neural network.

Important status
No preprocessing, route-grid generation, model training, or route validation has been completed yet. This guide documents the current files and the planned pipeline. Locations and coverage differ by dataset; rover samples and local HiRISE DTMs must not be mistaken for planet-wide layers.

Dataset root:
C:\Users\Harsh Goel\Downloads\MarsWalk\python\mars_datasets

1. Folder map

Keep source downloads unchanged in **data/raw**. Generated, aligned products go under **data/processed**. The route planner reads processed grids and metadata, not the original ZIP archives on every request.

```
mars_datasets/
|-- data/
|   |-- raw/                original downloads (keep unchanged)
|   |   |-- ai4mars/        rover images and terrain labels
|   |   |-- climate/        MCD_6.2.tar.gz (downloaded, not extracted)
|   |   |-- craters/        Robbins crater GeoJSON
|   |   |-- dem/            global blended HRSC/MOLA DEM GeoTIFF
|   |   |-- dunes/          equatorial, north-polar, south-polar archives
|   |   |-- geology/        SIM 3292 database ZIP
|   |   |-- ice/            SWIM products (currently empty/pending)
|   |   |-- radiation/      global image + Curiosity RAD sample
|   |   |-- terrain/        HiRISE local DTM (currently empty/pending)
|   |   |-- thermal_inertia/ TES binary/VICAR maps and previews
|   |   `-- weather/        Perseverance MEDA inventory and sol sample
|   |-- processed/
|   |   |-- rasters/        aligned numeric GeoTIFF/COG layers (empty)
|   |   |-- vectors/        normalized GeoPackage layers (empty)
|   |   |-- route_grid/     per-cell features / Parquet (empty)
|   |   `-- ai4mars/        image-mask manifests and splits (empty)
|   |-- catalog/            dataset_manifest.csv, factor_registry.csv
|   `-- exports/            small inspection tables only
|-- config/                 example vehicle, suit, mission profiles
|-- models/terrain_classifier/ AI4Mars segmentation model (empty)
|-- routes/results/         generated route results (empty)
`-- docs/                   this guide
```

Current generated outputs: the processed subfolders exist but have no processed layers. The two catalog CSVs exist; the dataset manifest tracks file/status information, while the factor registry currently has only its header. The three YAML example profiles contain null placeholders that must be replaced with validated mission values.

Catalog files

File	Purpose	Status
data/catalog/dataset_manifest.csv	Tracks dataset ID, path, byte size, download status, source and notes. Use it to audit what exists and what is pending.	Present; update after extraction/validation.
data/catalog/factor_registry.csv	Maps each route factor to unit, source product, preprocessing method, status and uncertainty.	Present but header only; populate during preprocessing.
README.txt	Explains directory conventions and current coverage limitations.	Present; update as work proceeds.
config/*.example.yaml	Vehicle, suit and mission constraints used in resource estimates and route feasibility.	Present; values intentionally null.

2. Global terrain and surface layers

These sources contribute map features used by routing. Raw products may use different projections, resolutions, vertical references, nodata codes and formats; preprocessing must reconcile those before combining values.

Layer	Current files	Contents / coverage	Use in planner	Preprocess before use
Global HRSC/MOLA DEM	data/raw/dem/Mars_HRSC_MOLA_BlendDEM_Global_200mp_v2.tif (+ .aux.xml)	11.38 GB; global blended elevation raster (~200 m product). Auxiliary XML accompanies it.	Elevation, derived slope/aspect/roughness, 3D terrain backdrop. Numeric raster; primary route cost input.	Inspect CRS, vertical units, nodata and affine transform; compute slope/aspect/roughness; create tiled/COG clips/resamples for route region.
Robbins crater catalogue	data/raw/craters/robbins_craters.geojson	Global crater FeatureCollection; previously parsed as 384,343 crater features.	Crater centers/diameters and available attributes; crater proximity/avoidance risk, map markers.	Validate coordinates/schema; derive spatial index and distance-to-crater or rim-proxy features. Do not claim rim height unless source has it.
SIM 3292 geology	data/raw/geology/sim3292_database.zip	790 MB ZIP database for global geologic map.	Geologic units/material context; categorical route-grid features and explanatory context.	Inspect archive; select authoritative geometry and attributes; reproject to Mars CRS; normalize to GeoPackage; rasterize categorical unit IDs when appropriate.
MGD3 dunes	data/raw/dunes/{equatorial,north_polar,south_polar}/	Equatorial GIS/GML/shapefile ZIPs; north-polar DVD archive; south-polar GIS archive plus small non-GIS package. Duplicate south-polar GIS ZIP retained under duplicates/.	Mapped dune boundaries; hazard proximity or categorical dune-field membership. Coverage is product-region based, not guaranteed complete hazard mapping.	Extract only once; inspect shapefile/GeoDatabase contents and CRS; deduplicate equivalent products; repair geometry if needed; compute distance/buffer and coverage flags.
TES thermal inertia	data/raw/thermal_inertia/tes_christensen_thermal_inertia_8ppd.vicar; tes_putzig_thermal_inertia_20ppd.bin; previews; SOURCE_NOTES.txt	Global/near-global TES-derived products; native grids and coverage differ. Preview images are visualization only.	Thermal inertia as a coarse surface-material proxy (e.g., rock/dust/sand context), never a direct wheel-slip or soil-strength measurement.	Read source labels/layout/projection and calibration; decode numerical raster; convert to GeoTIFF; align carefully; preserve source and uncertainty masks.
MCD 6.2	data/raw/climate/MCD_6.2.tar.gz	Full 4.93 GB archive now present. README says core files/software are bundled; optional dust/EUV scenarios live under addons. Not yet extracted or validated.	Modeled atmospheric conditions varying with location, season and local time; route-specific environmental context.	Review MCD README/terms; extract safely; install required runtime and use provided interface. Build query/cache by location, season, local time and scenario; record units and scenario IDs.

Resolution rule: the route grid should not pretend that coarse TES/MCD data have HiRISE-level detail. Store each feature's source resolution, valid-data coverage and confidence; sample/coarsen onto a selected planning grid without inventing finer information.

3. Learning data and environmental observations

Dataset/files	What is present	Contents / limitation	AI/planner use	Preprocess
AI4Mars merged	data/raw/ai4mars/ai4mars-dataset-merged-0.6.zip (16.23 GB)	Rover camera images plus human terrain labels/masks, according to the dataset package. Rover-view imagery is local and perspective-based, not a global orbital map.	Train/validate a terrain segmentation model (e.g., soil/sand/bedrock/rock classes provided by labels). Predictions can inform traversability only where image context exists.	Inspect archive and license; create image-mask manifest; verify paired files/classes; split by drive/scene/sol to reduce leakage; normalize images; train/validate/test; calibrate confidence. Model artifact belongs in models/terrain_classifier/.
MEDA weather sample	data/raw/weather/meda/inventories/docs and sample_sol_0001/ CSV/XML pairs	Inventory indexes a larger Perseverance archive. Downloaded actual files are a sample from sol 1 at Jezero. XML labels define columns/units. Not planet-wide weather.	Ground-truth/reference for local weather variability and checking MCD at Jezero; not a global raster or sufficient by itself to train global weather prediction.	Parse label metadata, timestamps/sols, units and quality flags; create tidy observation tables; map station location; compare with MCD only after matching time/location/variables.
MSL RAD sample	data/raw/radiation/msl_rad/RAD_RDR_2026_080_17_18_4843_V00.TXT, .LBL, RAD_RDR_DS.CAT	One Curiosity RAD RDR sample from Gale; label and catalog describe record structure. Not global surface-dose coverage.	Reference/validation sample for radiation scenario assumptions at Gale; cannot support direct per-cell global radiation training alone.	Read label; decode measurements and quality/time fields; retain units and calibration; join by location/time; keep observation provenance.
NASA Odyssey dose map	data/raw/radiation/global_mars_odyssey/PIA03480_global_mars_radiation.jpg	Global estimated dose visual downloaded as JPEG (227,062 bytes). Despite page link naming, local file bytes are JPEG; it lacks numeric georeferenced raster values.	Context image only; do not include as numerical route-grid dose or train a pixel model from its colors without a scientifically valid georeferencing/calibration source.	Keep as reference graphic; seek original scientific numeric product if a quantitative global radiation layer is needed.
HiRISE local DTM	data/raw/terrain/hirise/jezero/ (empty)	Jezero 1 m/pixel DTM identified but download was refused. HiRISE DTMs are local stereo products, not wall-to-wall global terrain.	High-resolution validation/planning only inside downloaded DTM footprints, once available with label/georeferencing.	Acquire IMG and matching PDS label; convert with PDS-aware GDAL/ISIS; validate coordinates and vertical values; make footprint mask; combine only as a local higher-resolution override.
SWIM ice	data/raw/ice/ (empty/pending)	SWIM product download failed due official endpoint timeouts. SWIM4MIM mapping is about 3 km pixels and roughly 60S-60N coverage; it does not cover the poles.	Optional subsurface-ice consistency/hazard context, not direct soil strength or definitive ice truth.	After acquisition, inspect metadata and product semantics; georeference/convert; preserve no-data and uncertainty; apply only inside the valid footprint.

Key distinction: rover records are measurements at rover tracks; global climate and orbital maps are modelled/mapped products. Label every feature as measured, modelled, derived, or missing, and never fill a missing global layer with a single rover sample.

4. Preprocessing and outputs to create

Process once when source data change. Route requests should read prepared grid tiles and cached climate values, not unpack multi-gigabyte archives at request time.

1 | Inventory & preserve

Check file size/hash, archive integrity, source version, license, coordinate system, nodata and units. Keep source archives read-only; update dataset_manifest.csv.

2 | Extract and decode

Extract ZIP/TAR/GDB products into staging folders. Read PDS labels, XML labels, VICAR labels and TES metadata. Convert native rasters/vectors with tools that honor map projection and scale.

3 | Normalize spatial layers

Choose a Mars coordinate reference system and cell size for each planning scale. Reproject vectors; align numeric rasters to a common extent/resolution; never treat preview PNG/JPEG as source values. Record valid-data masks and per-layer coverage.

4 | Derive physical features

From DEM derive elevation, slope, aspect, local roughness and relief. From vectors derive crater/dune proximity and geology class. From TES derive thermal inertia features. From MCD query/cache temperature, pressure, winds, density and relevant atmospheric fields at location/season/time.

5 | Build route-grid tiles

Create one row per grid cell per tile/scale, not one planet-sized CSV. Store cell ID, x/y or lat/lon, source feature columns, units, source IDs, valid-data flags and uncertainty. Prefer GeoTIFF/COG for rasters, GeoPackage for vectors and Parquet for tabular route cells.

6 | Prepare AI4Mars training set

Create linked image-mask manifest and group-aware train/validation/test splits. Train a vision segmentation network on rover images and masks. Export calibrated class probabilities; avoid claiming global generalization without orbital imagery labels and independent validation.

7 | Run route engine

A* evaluates neighboring cells using slope, terrain class probabilities, dune/crater/geology constraints, climate and mission-profile costs. Maintain fastest/safest/balanced weight sets. Route audit samples a user polyline against the same layers and reports segment risks and unknown-data flags.

8 | Validate and publish outputs

Validate on held-out labeled imagery and known rover traverses; check route costs, coordinate transforms, no-data behavior and edge cases. Return route geometry, distance/time/energy estimates, factor summaries, uncertainty/data coverage and explanations.

Suggested processed structure
processed/rasters: aligned DEM, slope, aspect, TES, climate caches, ice/radiation only when numeric source exists; processed/vectors: crater, dunes, geology GeoPackages; processed/route_grid: multi-resolution Parquet tiles plus schema/coverage metadata; processed/ai4mars: manifest, split lists, normalized masks and evaluation records.

5. What the AI will learn vs calculate

Component	Learns or calculates	Uses these data	Produces
Terrain vision model	Learns pixel-to-class patterns from labeled images; outputs class probabilities and confidence.	AI4Mars image/mask pairs.	Terrain segmentation on image pixels. Does not directly output a globally valid route.
Map feature pipeline	Calculates deterministic cell attributes and derived metrics.	DEM, crater/geology/dune vectors, TES, MCD and any properly georeferenced ice/radiation layers.	Per-cell numerical/categorical features with units, valid-data flags and uncertainty.
A* route planner	Searches a graph for a minimum-cost path under chosen constraints; it is an algorithm, not a trained neural model.	Route-grid cells + vehicle/suit/mission profile.	Fastest, safest, balanced candidate paths and cost breakdowns.
Route audit/explainer	Samples a user-drawn line and compares it with alternatives; creates factual explanation from computed metrics.	Same grid, constraints and route-cost model.	Segment hazards, high-risk cells, resource estimates, unknown-data notices and route comparison.

Example request/response contract

Input: start/end latitude-longitude, mission mode, vehicle/suit profile, season/local time or scenario, slope/slip/resource limits, and optional user polyline.
Output: route polylines, distance, duration estimate, max/mean slope, terrain/hazard segments, estimated energy and environmental budgets where supported, viability/cost score, data coverage, uncertainty, and an explanation of trade-offs.

What can be promised: computed routes over available map coverage, at the grid's true resolution, with provenance and data gaps shown. Do not promise exact oxygen/energy/slip or survival values until validated physics and vehicle parameters are supplied. A* gives the best route only relative to the chosen cost function and constraints.

Current readiness

Raw data collection is substantially underway and the MCD archive is present. Preprocessing is not done; the processed folders are empty; factor_registry.csv is unpopulated; example engineering profiles have nulls; no route model has been trained or validated. The immediate next step is safely inspect/extract MCD, then create geospatial preprocessing scripts and a small test-region route grid before attempting planetary scale.

6. Inventory notes and references

The following filenames were confirmed in the raw-data folder when this guide was prepared. The manifest is the detailed byte/status ledger. Archive-internal filenames will become visible after integrity checks and extraction.

Folder/category	Confirmed files
AI4Mars	ai4mars-dataset-merged-0.6.zip
DEM	Mars_HRSC_MOLA_BlendDEM_Global_200mp_v2.tif; .aux.xml
Craters	robbins_craters.geojson
Geology	sim3292_database.zip
Dunes	equatorial: Geodatabase.zip, GML.zip, mgd3_equatorial_shapefiles.zip, Shapefiles.zip; north_polar: mgd3_north_polar_dvd.zip; south_polar: mgd3_south_polar_gis.zip, SP_DataForNon-GIS-Users.zip; duplicates/SP_GIS_Download_Package.zip
Climate	MCD_6.2.tar.gz (4,930,599,406 bytes; downloaded, not extracted)
TES	tes_christensen_thermal_inertia_8ppd.vicar; tes_christensen_thermal_inertia_preview.png; tes_putzig_thermal_inertia_20ppd.bin; tes_putzig_thermal_inertia_preview.jpg; SOURCE_NOTES.txt
MEDA	meda_bundle_readme.txt; meda_calibrated_env_inventory.csv; meda_derived_env_inventory.csv; meda_env_rdr_sis.pdf; sample_sol_0001/ four CSV/XML product pairs
Radiation	global_mars_odyssey/PIA03480_global_mars_radiation.jpg; msl_rad/RAD_RDR_2026_080_17_18_4843_V00.TXT and .LBL; RAD_RDR_DS.CAT
Pending/empty	SWIM ice products; Jezero HiRISE DTM. The corresponding raw folders are currently empty.

Primary references

MCD 6.2 request and software package: https://www-mars.lmd.jussieu.fr/MCD_pro/mcd_pro.html

MCD web interface and data-access notes: https://www-mars.lmd.jussieu.fr/mcd_python/index.html

MCD user manual (6.2 software package): https://www-mars.lmd.jussieu.fr/mars/info_web/user_manual_4.3.pdf

SWIM4MIM products and coverage notes: https://swim.psi.edu/SWIM4MIMProducts.php

HiRISE Jezero DTM product page: https://hirise.lpl.arizona.edu/dtm/ESP_048842_1985

NASA Mars Odyssey global radiation map: https://science.nasa.gov/photojournal/estimated-radiation-dosage-on-mars/

PDS MEDA bundle: https://pds.nasa.gov/ds-view/pds/viewBundle.jsp?identifier=urn%3Anasa%3Apds%3Amars2020_meda&version;=1.0

PDS MSL RAD dataset: https://pds.nasa.gov/ds-view/pds/viewDataset.jsp?dsid=MSL-M-RAD-3-RDR-V1.0

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